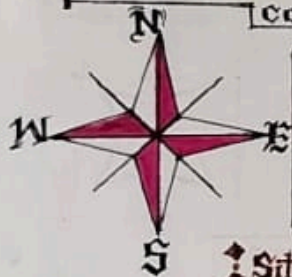


SITE ANALYSIS & CONTEXT



Design Vision & Challenge

The 2050 Climate Crisis: By the year 2050, accelerating sea-level rise and unpredictable marine storms pose an existential threat to low-lying coastal topographies like the Maldives. Traditional rigid coastal architecture is highly vulnerable to sinking, structural displacement, and the catastrophic reef destruction.



The Architectural Imperative: To survive, humanity must shift from fighting the ocean to living with it. This project introduces a self-sustaining amphibious typology that respond dynamically to oceanic forces, utilizing natural ecosystem to transition man-made structures into living, resilient habitats.

Site Context & Environmental Notes

Location & Geography: Situated on the eastern reef rim of Kooju Atoll, Maldives, at a low-lying elevation of just 1.5m above sea level. This highly vulnerable site is directly exposed to open ocean swells, serving as a critical testing ground for 2050 resilient amphibious design.

Location: Climatic Condition - Marine tropical climate with humidity, constant solar radiation & monsoon wind.

Ecological Context: Surrounded by sensitive living coral reefs that act as natural strict wave breakers.

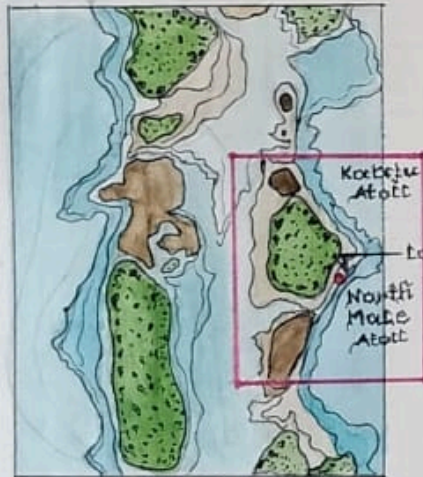
Site-Responsive Engineering:

Hydrostatic Stability:

$$F_b = F_g + F_t$$

Self-Righting: Axis alignment (B over G)

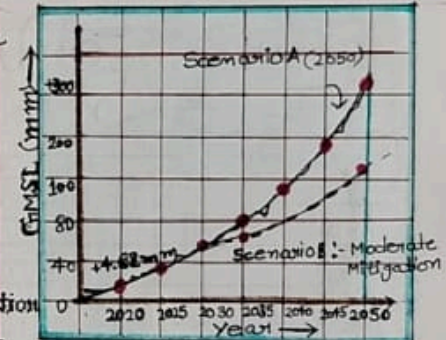
Tethered Anchoring:
 Sealed cable tension.



Maldives atoll map



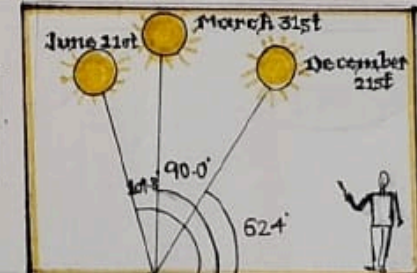
Micro site location & Reef Geomorphology



Projected Sea Level Rise & Vulnerability (2026-2050)



Solar Angles & Radiation (Maldives)



THE AMPHIBIOUS ECO-POD

Design Concept & Story

Form evolution

• Residential Pods
Organic Atoll Zoning

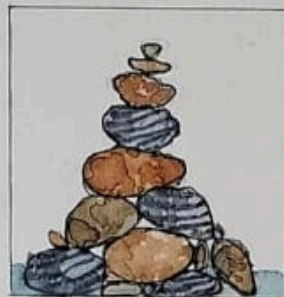
• Manta Wings
Hydrodynamic Geometry
reduces underwater drag.

The Narrative

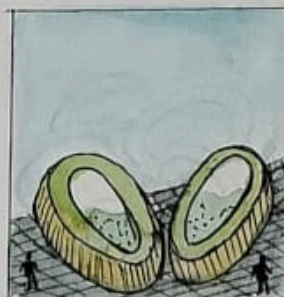
Imagine waking up in 2050 in a home that responds to the rhythm of the Ocean-floating seamlessly on calm days submerging during extreme to extreme marine storms to protect its occupants.

The Execution

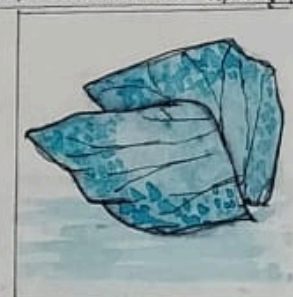
A smart, semi-submersible pod engineered with an 80% lightweight carbon-fiber skeleton and 20% porous external panels left open for living coral ecosystems to grow and design their natural art.



• Stone Stacking
Inspired by stone balancing to optimize the pod's center of gravity buoyancy stability.



• Coral (Bio-Regenerative)
Living eco-skeleton walls.

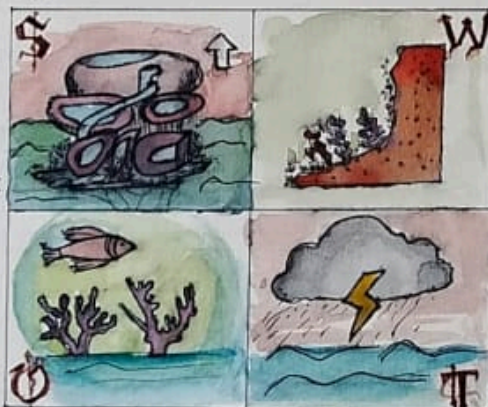


S Strength - High hydro-dynamic stability & perfect buoyancy.

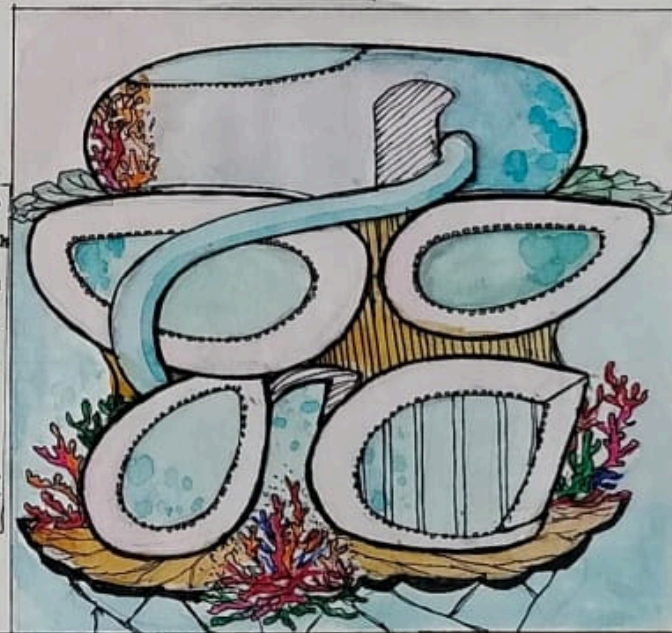
W Weakness - Structural dependency during initial coral growth phase.

O Opportunity - Ecological restoration through marine habitat regeneration.

T Threat - Vulnerability to unpredictable 2050 extreme storms.



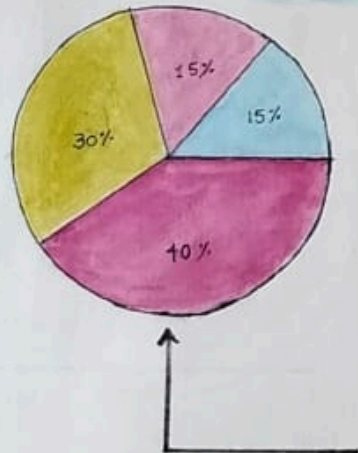
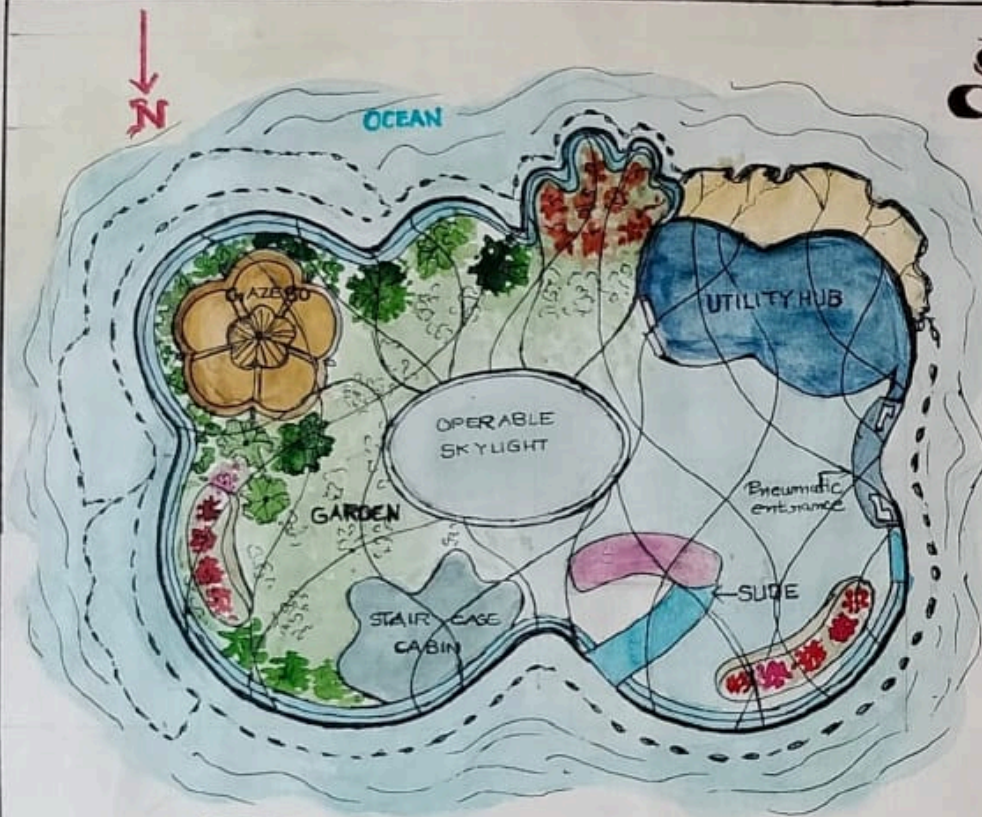
A design that stop fighting the ocean and start living with it. A man-made structure into a living, breathing part coral ecosystem.



Anchorage System -
Tethered to the ocean floor via high-tension mooring cables, allowing dynamic flotation.

N.T.S

SITE PLAN & STATS



Space Programming Analysis

- Private Living Zone (40%) : (144° angle)
- Semi-Public & Common Areas (30%) : 108° angle
- Service & Utilities (15%) : (54° angle)
- Circulation & Special Features (15%) : 54° angle

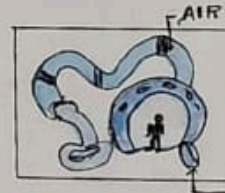
Building Material



- Carbon Fiber: High-Strength, Ultra-lightweight Shell.



- PMMA Sheet: Transparent, impact-resistant glazing.



- Pneumatic Core: Air-filled structural entrance.



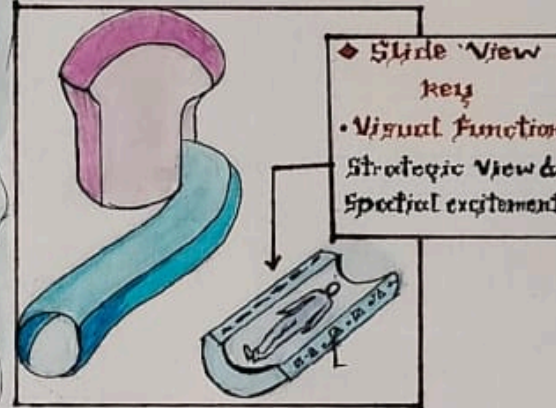
- Technical Wood: Water-resistant engineering timber.



- Microalgae Panels: Natural oxygen-generating facade.

Site Statistics

- Site Location: Kaahe Atoll, Maldives (Offshore).
- Project Footprint: ≈ 140 Sq.m (Organic Envelope).
- Spatial Envelope: 18m x 12m Max Boundaries.
- Virtual Anchor Diameter: $\varnothing 20$ m
- Water Depth: 15m - 20m Average.
- Mooring Type: Cable-Tethered.



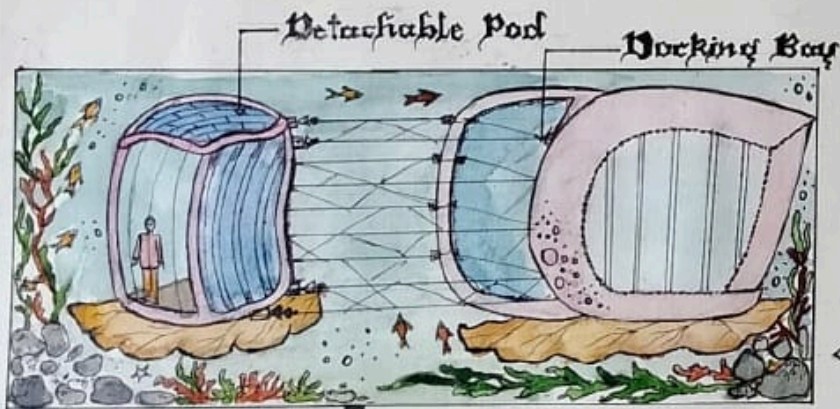
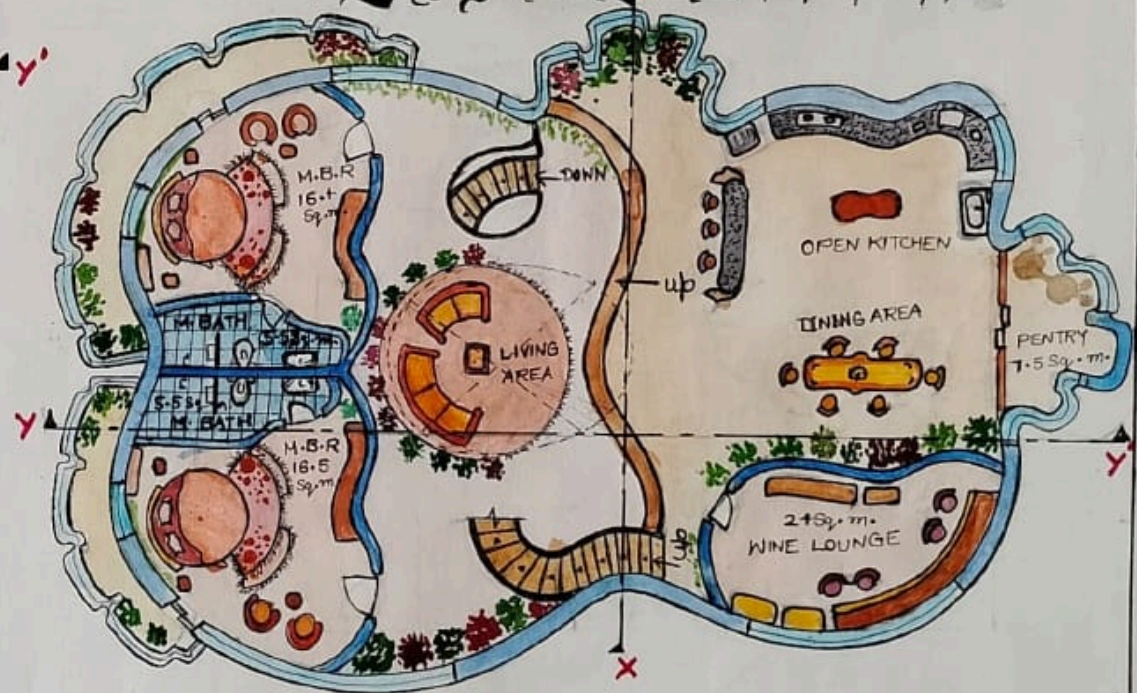
- Slide View key
- Visual Function: Strategic View & Special excitement

FLOOR PLANS

GROUND FLOOR



FIRST FLOOR



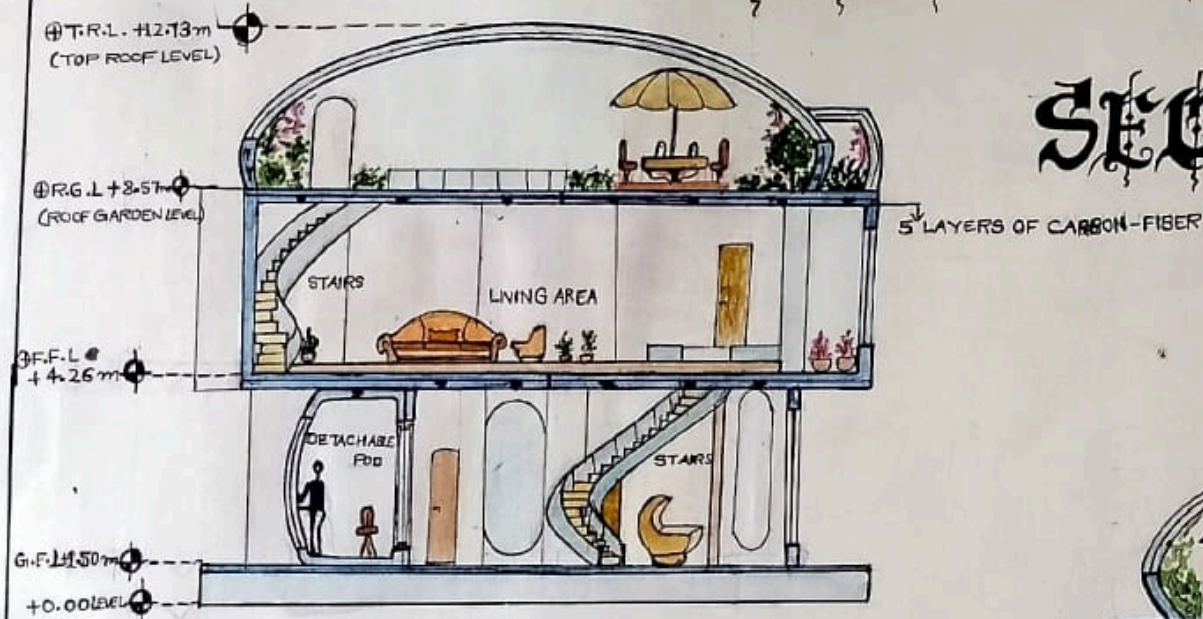
Perspective View

SCALE - 1:100

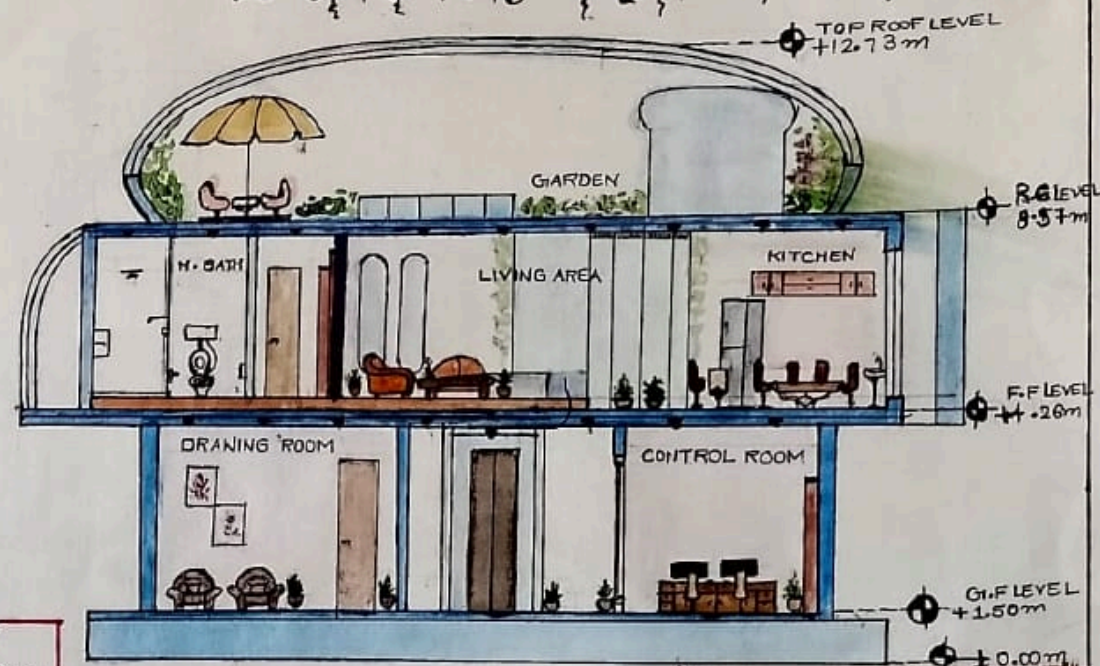
TEAMID - TUH-A-I-GRP-0017
SHEET NO - 4

SECTIONS

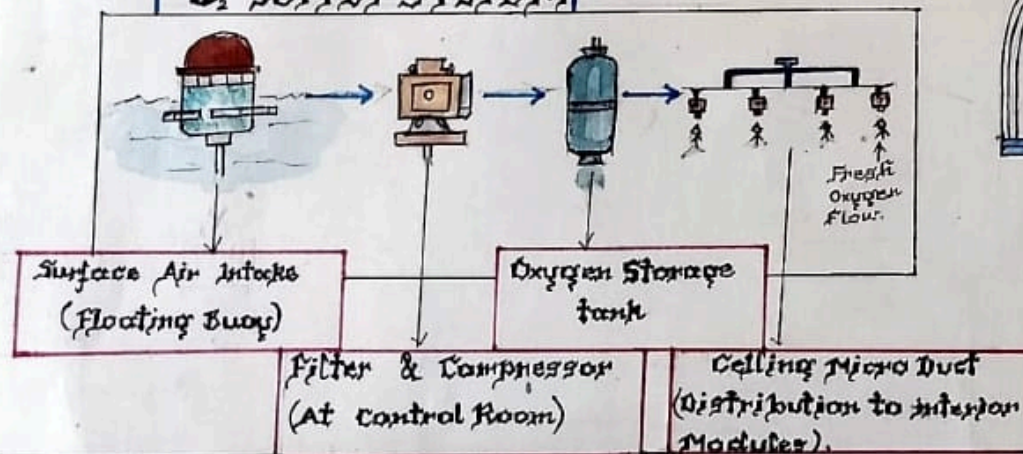
SECTION X-X'



SECTION Y-Y'



O₂ SUPPLY SYSTEM

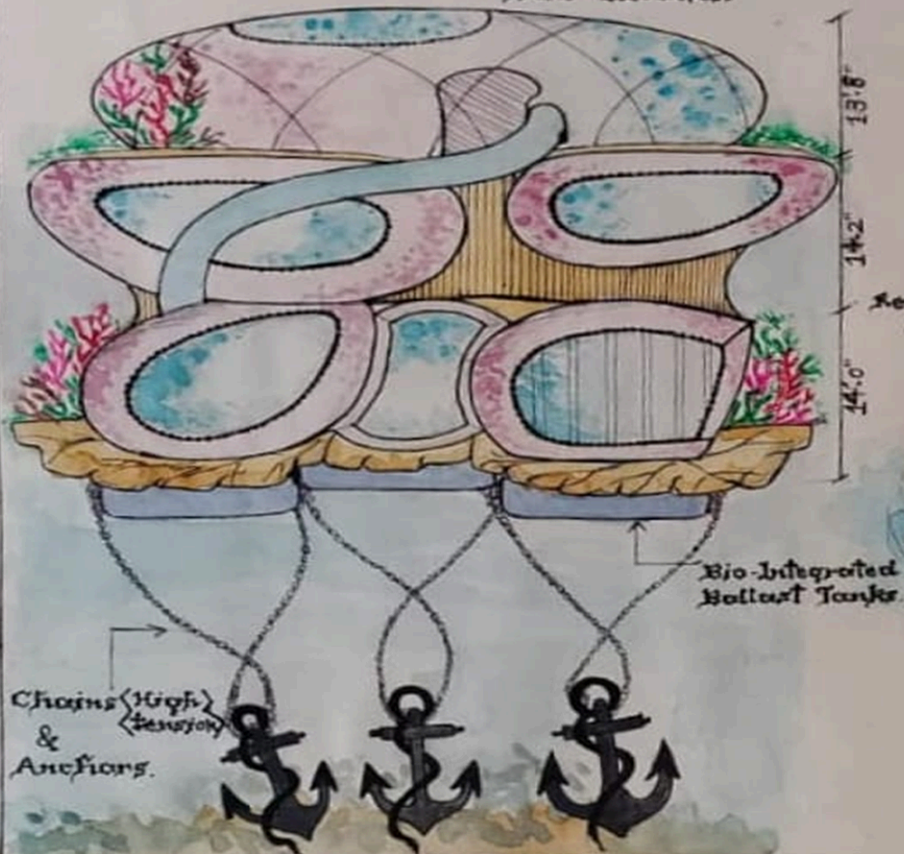


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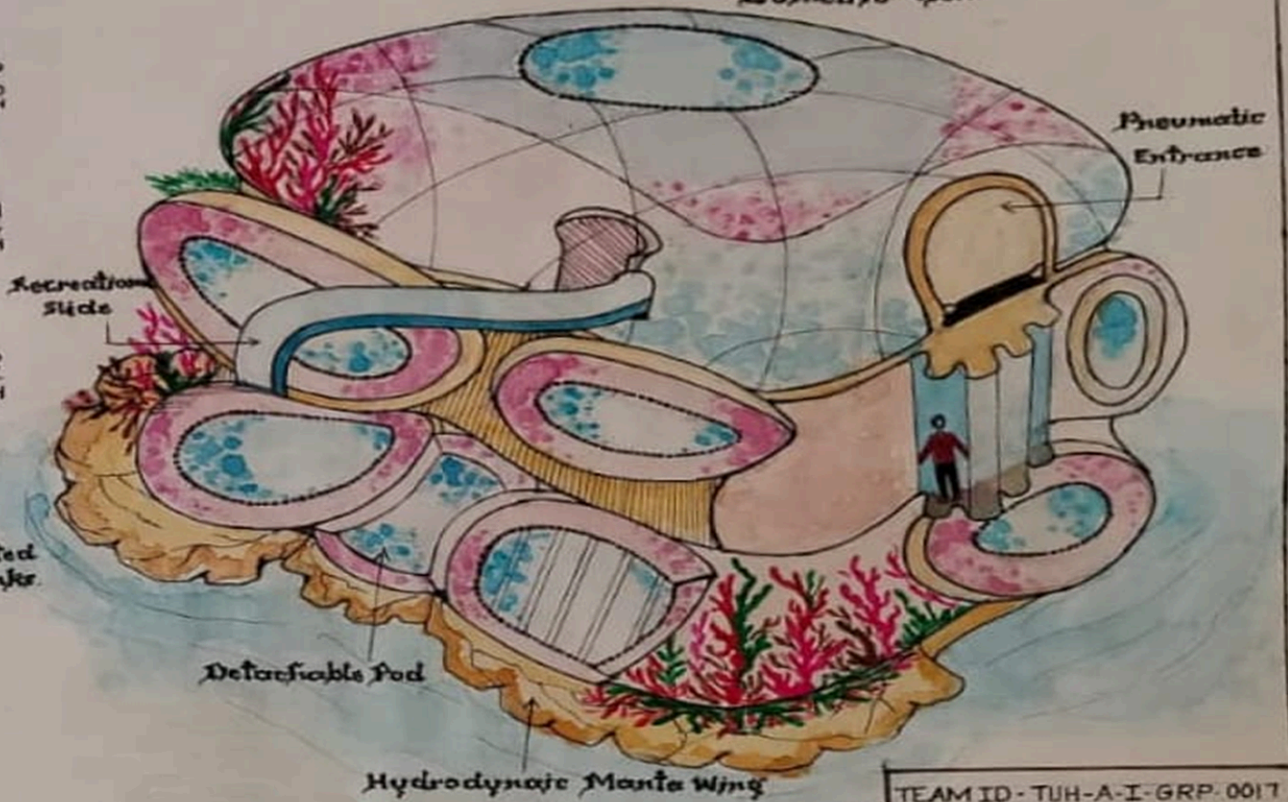
TEAM ID - TUH-A-I-GRP-0017
SHEET NO - 5

ELEVATION & VIEW

"Front Elevation"



"Isometric View"



TEAM ID - TUH-A-I-GRP-0017
SHEET NO. - 6

